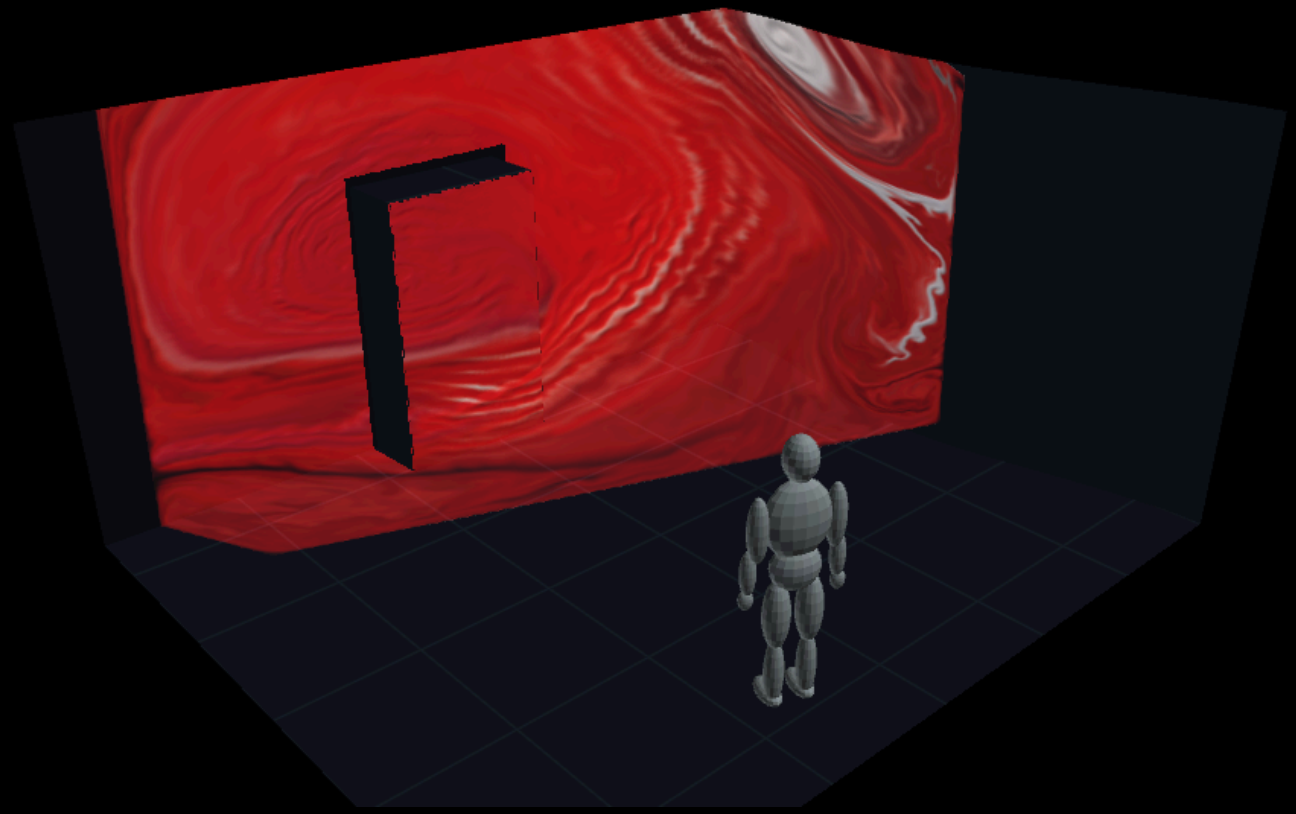
PROTOTYPE PROGRESS

Anamorphic Fluid Room

A room-scale image that responds to voice and keeps moving through silence.

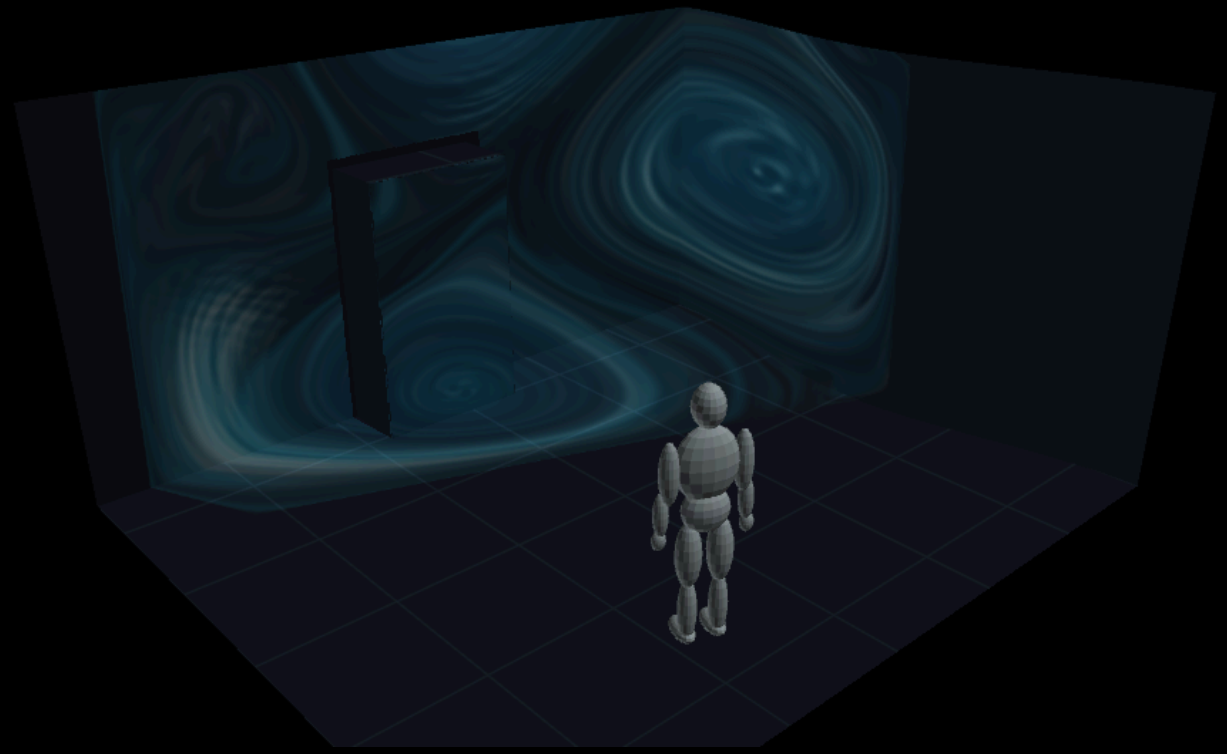


Intensity palette audition in TouchDesigner

16 October 2026

The virtual room is working

- **6 × 5 × 3 metre room** with a sharp corner, inward curve and protrusion.
- **Movable mannequin** with eye-level and orbit inspection.
- **Live microphone input** changes the fluid and its palette.
- **Editable Blender asset** accompanies the native TD project.

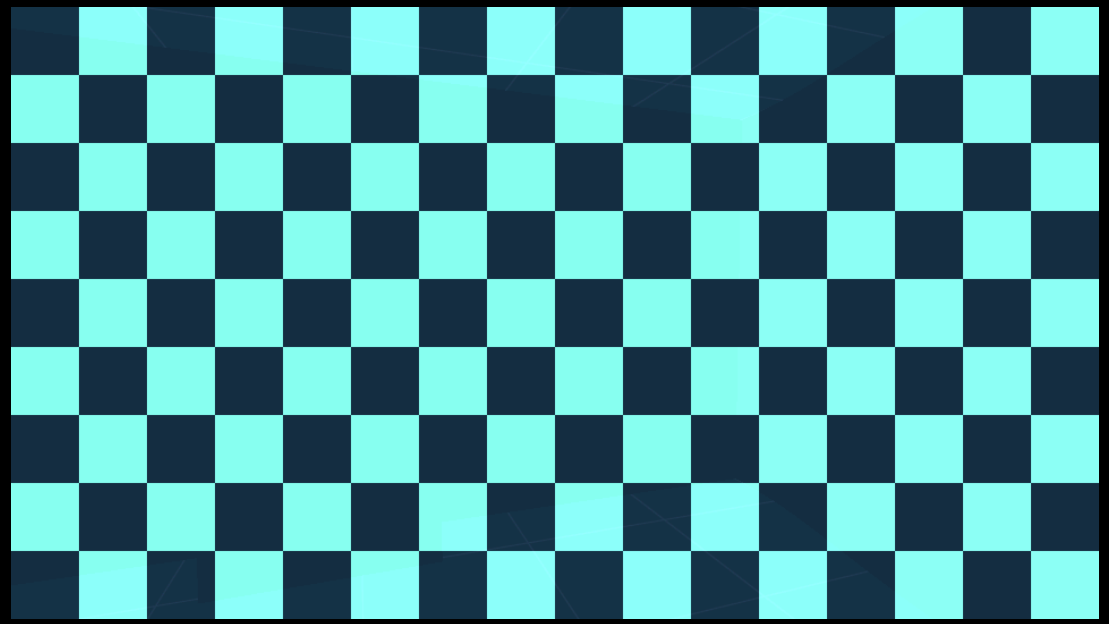


Third-person cutaway reveals the geometry and viewing position.

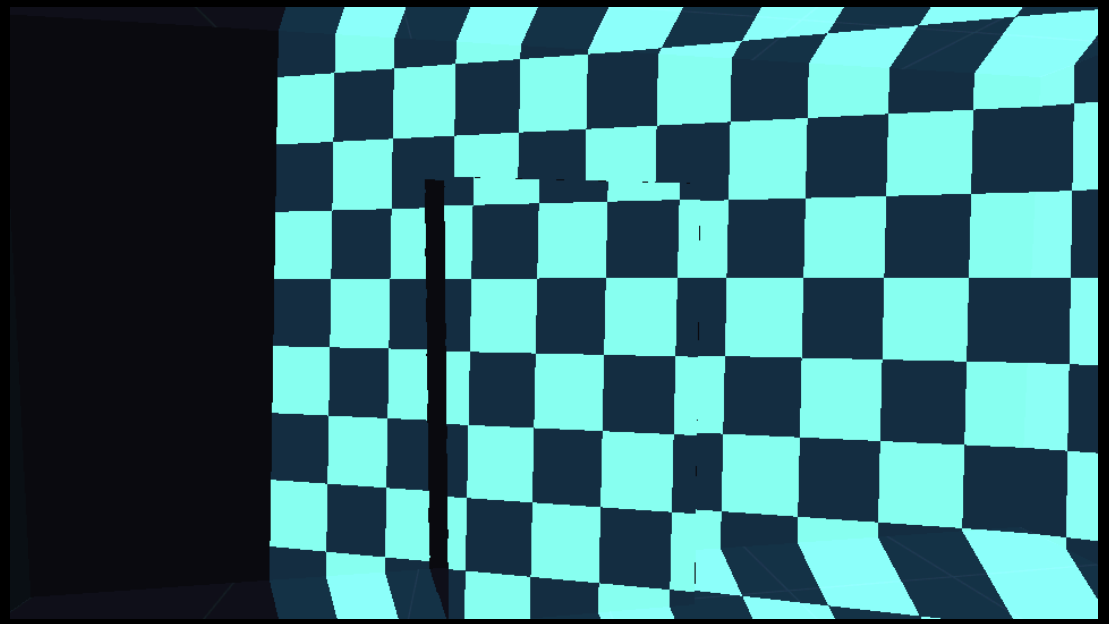
The current prototype previews ideal mapped imagery. Physical projector calibration comes later.

Alignment depends on the sweet spot

One fixed projection camera aligns the image across uneven surfaces.



At the sweet spot
The checkerboard reads as a continuous grid.



After moving away
The room geometry becomes visible through distortion.

Moving the person leaves projection alignment fixed. “Set sweet spot here” re-anchors the image without resetting the fluid.

The audio-to-image pipeline

STEP	STAGE	WHAT IT CONTRIBUTES
01	**Microphone analysis**	RMS loudness and confidence-filtered fundamental pitch.
02	**Emotional mapping**	Continuous weights for Calm, Wonder, Joy and Intensity.
03	**GPU fluid simulation**	Voice adds force and ink to persistent idle circulation.
04	**Colour mapping**	The palette changes across existing pigment as well as new ink.
05	**Room projection**	The sweet spot aligns imagery. Independent cameras inspect it.

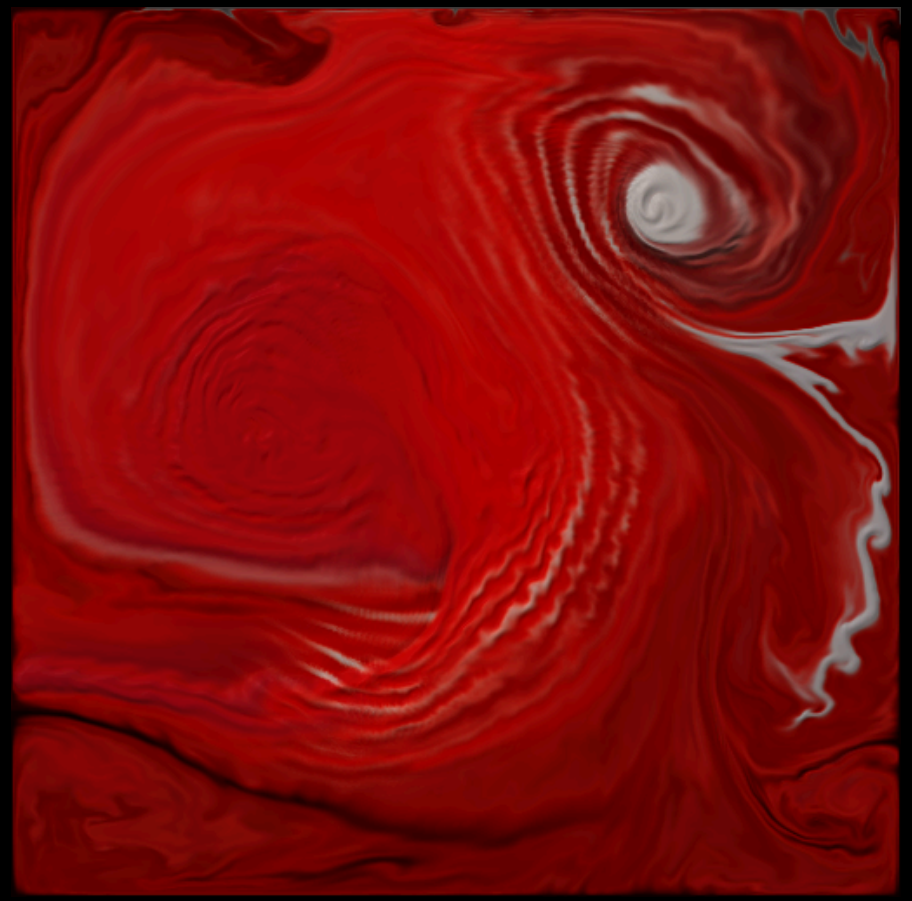
TouchDesigner embeds the shaders, Python callbacks and interface in one working TOE file.

Mood follows the character of the voice

ARTISTIC MOOD	ACOUSTIC TENDENCY	PALETTE DIRECTION	MOVEMENT DIRECTION
Calm	Quiet, lower pitch	Deep blue and teal	Gentle circulation
Wonder	Quiet, higher pitch	Indigo and lavender	Delicate curls
Joy	Louder, higher pitch	Coral and gold	Faster swirls
Intensity	Louder, lower pitch	Burgundy and red	Denser streams

These names describe the artwork’s expression. They do not identify the speaker’s emotions. Neighbouring moods blend continuously.

The fluid now holds finer detail



Refined solver capture in Intensity audition.

Recent improvements

- Curl forces help retain rolling eddies.
- Corrected pigment transport preserves thin trails.
- Pale ink folds into coloured pigment.
- Viscosity stays separate from drag and colour fading.

Measured prototype performance

≈60 FPS

512 × 512 fluid on an RTX 4060 Laptop GPU

VIEW	AVERAGE	MINIMUM*
Comparison	60.02	59.32
First person	60.08	59.95

Recorded checks

- **65.6 seconds** of below-gate idle with finite, bounded fluid data.
- **Tone tests** resolve within one semitone across the tested speech range.
- **Higher viscosity** reduces velocity-field variation by about 29%.
- **Camera movement** preserves the fixed projection anchor.

Recorded 13 Oct 2026 in TouchDesigner 2025.33230. *Minimum sampled, smoothed project FPS. These short runs do not establish installation endurance or performance with physical projector outputs.

Next checks on a real site

01 **Measured room geometry**

Replace the example room with a clean, dimensioned mesh and agree on the intended viewing position.

02 **Voices in the actual space**

Calibrate the noise floor, try different speakers and test how room acoustics affect pitch confidence.

03 **Projector coverage and calibration**

Check throw, surface brightness, overlapping coverage and the shadows cast by visitors.

04 **Installation endurance**

Confirm a cold reopen, microphone recovery and longer unattended operation with the final output setup.

Physical projection and real-person tracking remain outside this virtual prototype.

Live Demo